Session 2.2: Business Data Visualization: Exploratory Plotting and Executive Dashboards

TODAY'S PURPOSE

Why This Session Matters

- Turn data tables into clear charts
- Match each chart type to its business question
- Build a dashboard executives read at a glance

Today's flow: 60 min lecture, then 120 min hands-on lab

EXPECTED OUTCOMES

By the end of today, you will be able to:

1. Choose charts for trends, categories, and relationships
2. Build line, bar, and heatmap charts in Matplotlib
3. Export a 2x2 dashboard and fix an IndexError

Concepts for This Session

CONCEPT 1

Visual Encoding Hierarchy

Mapping quantitative commercial data to spatial coordinates, lengths, and colors following cognitive visual perception principles.

CONCEPT 2

Figure & Axes Architecture

The object-oriented Matplotlib canvas model where a top-level Figure container manages one or more individual Axes plotting subplots.

CONCEPT 3

Time-Series Revenue Curves

Continuous line plots visualizing revenue, marketing spend, and customer acquisition costs across fiscal months.

CONCEPT 4

Categorical Distribution Plots

Clean bar plots and box plots created with Seaborn to compare performance metrics across corporate business units.

How the Concept Works in Practice

FOUNDATION 1

Boardroom Visual Hygiene

Executive leadership teams make multi-million dollar decisions based on quantitative charts. Visualizations must eliminate chartjunk, 3D effects, and misleading truncated axes, prioritizing clear data ink ratios.

FOUNDATION 2

Object-Oriented Matplotlib Hierarchy

Avoid procedural plotting (plt.plot()) in favor of explicit object-oriented architecture (fig, ax = plt.subplots()). This enables precise control over tick marks, dual y-axes, titles, and export resolutions.

FOUNDATION 3

Statistical Density & Outlier Detection

Using distribution visualizations (box plots, histograms) to identify revenue skews and extreme transaction outliers that distort simple arithmetic averages.

FOUNDATION 4

Cognitive Visual Bandwidth

Selecting harmonious, accessible palettes (such as Harvard Crimson #991B1B and Slate #475569) with clear high-contrast typography, ensuring dashboards remain legible when printed in grayscale.

Build, Test, and Diagnose in the IDE

[1] FILE EXPLORER

Organize Assets

Use the session dataset and keep scripts in a clear workspace.

[2] CODE EDITOR

Write Small Segments

Run each completed segment before expanding the program.

[3] AI AGENT CHAT

Diagnose Errors

Capture the traceback, request an explanation, and review the fix.

[4] INTEGRATED TERMINAL

Verify Execution

Confirm the final program completes with return code 0.

Script Demonstration 1: Business Data Visualization: Exploratory Plotting and Executive Dashboards

```
import matplotlib.pyplot as plt
    import pandas as pd

    df = pd.read_csv("data/session_02_02_dataset.csv")

    fig, ax1 = plt.subplots(figsize=(10, 5))
    ax1.plot(df["Month_Name"], df["Organic_Revenue_USD_M"],
color="#991B1B", marker="o", label="Revenue (USD M)")
    ax1.set_ylabel("Organic Revenue (USD M)", color="#991B1B")

    ax2 = ax1.twinx()
    ax2.plot(df["Month_Name"], df["Customer_Acquisition_Cost_USD"],
color="#475569", linestyle="--", label="CAC (USD)")
    ax2.set_ylabel("Customer Acquisition Cost (USD)", color="#475569")
    plt.title("Revenue Growth vs. Acquisition Cost Efficiency")
    plt.savefig("data/dual_axis_trend.png", dpi=150,
bbox_inches="tight")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 2: Business Data Visualization: Exploratory Plotting and Executive Dashboards

```
import seaborn as sns

    fig, ax = plt.subplots(figsize=(8, 4))
    sns.barplot(data=df, x="Month_Name",
y="Paid_Marketing_Spend_USD_M", color="#991B1B", ax=ax)
    ax.set_title("Paid Marketing Budget Allocation by Month")
    ax.set_ylabel("Marketing Spend (USD M)")
    plt.savefig("data/marketing_spend_bar.png", dpi=150,
bbox_inches="tight")
```

SYNTAX

Readable Structure

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Run the example in the integrated terminal and inspect the output.

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Script Demonstration 3: Business Data Visualization: Exploratory Plotting and Executive Dashboards

```
numeric_cols = ["Organic_Revenue_USD_M", "Paid_Marketing_Spend_USD_M",
"Customer_Acquisition_Cost_USD", "Net_Promoter_Score"]
    corr_matrix = df[numeric_cols].corr()

    fig, ax = plt.subplots(figsize=(6, 5))
    sns.heatmap(corr_matrix, annot=True, cmap="Reds", fmt=".2f",
ax=ax)
    ax.set_title("Commercial KPI Correlation Matrix")
    plt.savefig("data/kpi_correlation_heatmap.png", dpi=150,
bbox_inches="tight")
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

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Exercise 1 and Exercise 2

EXERCISE 1: GUIDED

Foundational Practice

Follow the lab protocol, run the script, and verify its output with the session dataset.

EXERCISE 2: DEBUGGING

Applied Challenge

Extend the program, diagnose an intentional error, apply the verified correction, and confirm return code 0.